SAILAJA V MADHURAPANTULA

M.S. (Health Data Science), M.B.A. (Hosp. Administration)
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SUMMARY

- Data Analyst with over 12 years of combined clinical and analytical experience, including 7 years in healthcare analytics and 2 years of U.S. work experience.
- Proficient in SQL (BigQuery, Snowflake), Python, R, Tableau, Power BI, and Google Cloud Platform (GCP).
- Experienced in predictive modeling, EHR analytics, and clinical quality reporting.
- Skilled in the full analytics lifecycle from data ingestion and transformation to modeling, visualization, and performance reporting using large, cloud-based healthcare datasets and EHR systems.
- Published research work in high-impact international journals, recognized for independent analysis and precise documentation.
- Proven ability to leverage data to drive decisions and improve healthcare outcomes.

TECHNICAL SKILLS

Data Analytics & Engineering: SQL (BigQuery, Snowflake), Python (Pandas, NumPy, Matplotlib, Scikit-learn, Seaborn, Plotly, Dask), R, SAS; ETL/ELT, CTEs, Stored Procedures, Window Functions, Data Wrangling, Statistical Modeling, EDA, Machine Learning, Predictive Analytics
Visualization & BI: Tableau, Power BI, Looker Studio; Executive Dashboards, Reporting, Data Summarization, Business Insights
Microsoft Excel: Pivot Tables, VLOOKUP, Advanced Formulas, Data Cleaning, Reporting, and Visualization
Data Quality & Compliance: Data Validation, Quality Assurance, Maintaining Data Integrity
Healthcare Analytics: Clinical Quality & Patient Safety, EHR Data, Operational Analysis
Technical Tools & Support: Cloud Platforms (GCP, Snowflake), IT Coordination
Communication & Collaboration: Translating Complex Data Findings for Non-Technical Stakeholders, Cross-Functional Collaboration, Stakeholder Engagement

WORK EXPERIENCE

Intern - Acute Care Quality Data Analytics | Ascension | St. Louis, MO **05/2025 – 08/2025**

- Conducted predictive analysis for the Hospital Acquired Condition Reduction Program (HACRP) across more than 40 hospitals, applying CMS scoring methodology for FY 2026 through 2028 to identify hospitals with the highest penalty exposure, improving intervention targeting by 30%.
- Designed and optimized CTE-based SQL pipelines (ELT) in BigQuery to process NHSN and CMS datasets for hospital-level penalty scoring.
- Developed interactive Tableau dashboards visualizing “countdown to penalty” metrics, allowable event thresholds, and risk trends, enabling real-time decision making by clinical and executive teams.
- Engineered scalable architecture for real-time monitoring of hospital-acquired infections and penalties, enabling proactive interventions.

Graduate Research Assistant | Heartland Center | St. Louis, MO **07/2024 - 07/2025**

- Implemented a classification and tagging system for 300 LMS course modules, enhancing content discoverability and learner experience.

- Supported planning, execution, and follow-up for 4 live webinars featuring speakers from state and regional public health departments, strengthening engagement with local public health professionals.
- Optimized LMS usability, reducing navigation time by 20% and improving workflow efficiency.
- Documented cross-functional training initiatives, ensuring timely deliverables and consistent communication.

Graduate Researcher | Zhengzhou University | Zhengzhou, Henan, China **09/2018 – 07/2023**

- Published 3 peer-reviewed research studies in high-impact international journals, including one cited 27 times.
- Conducted multi-cohort statistical analysis on 1,000 to 10,000+ participant datasets using R, Python, and SQL.
- Led data wrangling, feature engineering, and modeling for cross-sectional and meta-analytical studies.
- Collaborated with clinicians and researchers on study design, data interpretation, and publication.

PROJECTS

High-Performance Computing for Analyzing Motor Vehicle Collision Trends **04/2025**

- Analyzed over 1.7M NYC collision records using Dask parallelization, reducing execution time by 62%.
- Built Random Forest models that achieve 99.93% accuracy to predict injury likelihood based on location, vehicle type, and contributing factors.
- Delivered actionable insights for public safety policy and high-risk area identification.

Predictive Modeling for Intensive Care Unit Patient Outcomes **12/2024**

- Developed machine learning models to predict ICU patient outcomes using the MIMIC-IV dataset.
- Conducted EDA, feature selection, and statistical analysis on patient demographics, vital signs, and length of stay.
- Applied K-Nearest Neighbors (KNN), Naïve Bayes, Support Vector Machine (SVM), and Random Forest, achieving 82.85% accuracy with Random Forest.
- Identified key predictors of mortality and ICU readmission, aiding clinical decision-making.

Diagnosis-Related Groups (DRGs) and Hospital Coverage Analysis **04/2024**

- Analyzed Washington state inpatient/outpatient data to identify DRG coverage gaps and revenue opportunities.
- Used R Studio for statistical tests, correlation analysis, and regression modeling ($R^2 = 0.97$) to optimize hospital resource allocation and revenue strategies.

Exploratory Data Analysis (EDA) of Gastric Cancer Biomarkers **12/2023**

- Conducted exploratory data analysis to identify potential biomarkers; used Python for data wrangling, statistical analysis, and visualization.

EDUCATION

M.S. Health Data Science | Saint Louis University | St. Louis, MO, USA | GPA 3.8/4.0
MBA Hospital Administration | Andhra University | India | GPA 3.2/4.0
B.S. Physical Therapy | Dr NTR University of Health Sciences | India | GPA 3.0/4.0

PUBLICATIONS

3 peer-reviewed publications in international journals (ORCID).

AWARDS

- Presidential & CSC Scholarship, Zhengzhou University (2018).